HDOC Day-11 Activity

Question: Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum of squares of its digits using a while loop.
- Find the sum of cubes of its digits using a for loop.
- Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop.

Step 1: Understanding the Problem

Input: A positive integer n and a menu choice (1, 2, or 3).

Output: The result of the selected digit operation.

Logic: Extract each digit using digit = temp % 10 and remove it using temp = temp / 10.

Choice 1: Sum of squares = Σ(digit × digit), using a while loop.

Choice 2: Sum of cubes = Σ(digit × digit × digit), using a for loop.

Choice 3: Difference = sum of cubes - sum of squares, using a do-while loop.

Step 2: Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n=123, choice=1	Sum of squares = 14	Sum of squares = 14	Pass
2	n=123, choice=2	Sum of cubes = 36	Sum of cubes = 36	Pass
3	n=123, choice=3	Difference = 22	Difference = 22	Pass
4	n=45, choice=1	Sum of squares = 41	Sum of squares = 41	Pass
5	n=45, choice=2	Sum of cubes = 189	Sum of cubes = 189	Pass
6	n=45, choice=3	Difference = 148	Difference = 148	Pass

Step 3: Algorithm and Evaluation

1. Start.

2. Declare n, temp, digit, choice, sumSquare, sumCube and difference as integers.

3. Read a positive integer n.

4. Display the menu: 1. Sum of squares, 2. Sum of cubes, 3. Difference between cube-sum and square-sum.

5. Read choice.

6. Use switch(choice).

7. For case 1: set temp=n and sumSquare=0. While temp>0, extract digit, add digit*digit to sumSquare, and remove the digit. Display sumSquare.

8. For case 2: set temp=n and sumCube=0. Use a for loop while temp>0; extract each digit, add digit*digit*digit to sumCube, and remove the digit. Display sumCube.

9. For case 3: set temp=n, sumSquare=0 and sumCube=0. Use a do-while loop to calculate both sums. Compute difference=sumCube-sumSquare and display it.

10. For any other choice, display Invalid choice.

11. Stop.

Evaluation: For 123, the square-sum is $1^2+2^2+3^2=14$, the cube-sum is $1^3+2^3+3^3=36$, and the required difference is 36-14=22. The prepared test cases therefore match the expected outputs.

Step 4: C Programming

```
#include <stdio.h>

int main()
{
    int n, temp, digit, choice;
    int sumSquare = 0, sumCube = 0, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("1. Sum of squares of digits\n");
    printf("2. Sum of cubes of digits\n");
    printf("3. Difference (cube sum - square sum)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0) {
                digit = temp % 10;
                sumSquare += digit * digit;
                temp /= 10;
            }
            printf("Sum of squares = %d\n", sumSquare);
            break;

        case 2:
            for (temp = n; temp > 0; temp /= 10) {
                digit = temp % 10;
                sumCube += digit * digit * digit;
            }
            printf("Sum of cubes = %d\n", sumCube);
            break;

        case 3:
            temp = n;
            do {
                digit = temp % 10;
                sumSquare += digit * digit;
                sumCube += digit * digit * digit;
                temp /= 10;
            } while (temp > 0);
            difference = sumCube - sumSquare;
            printf("Difference = %d\n", difference);
            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
```

```
}
```

Step 5: Compilation and Execution

The program was compiled and executed in Kali Linux and evaluated against the prepared test cases.

Step 6: Kali Linux Compilation and Execution

```
(kali㉿kali)-[~]
$ nano day11.c
(kali㉿kali)-[~]
$ gcc day11.c -o day11
(kali㉿kali)-[~]
$ ./day11
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cube sum - square sum)
Enter your choice: 1
Sum of squares = 14
```

Test Case 2:

```
(kali㉿kali)-[~]
$ ./day11
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cube sum - square sum)
Enter your choice: 2
Sum of cubes = 36
```

Test Case 3:

```
(kali㉿kali)-[~]
$ ./day11
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cube sum - square sum)
Enter your choice: 3
Difference = 22
```